Namasivaayam L

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Portfolio GitHub Linkedin

Professional Summary

Machine Learning Engineer specializing in **Generative AI**, **Multi-Modal Systems**, and full-stack development — currently shipping production GenAI for financial document processing at **Rocket**. Recent work: a modular **decision graph** that lifted reviewer-agreed accuracy from **29% to 77% (2.7x)** and cut wrong classifications by **95%** across ∼**17,500** weekly inbound items. Also took document extraction accuracy from **9% to 90%** with a production Gemini pipeline, and led the end-to-end ML development of an **LLM validation framework** that benchmarks in-production models against top market competitors. Experienced in building **HITL, RAG, & cloud-native deployments**.

Technical Skills

AI/ML: Deep Agents, LangChain, LangGraph, RAG, MCP, PyTorch, TensorFlow, Google ADK

Certifications: AWS certified Generative AI Developer - Professional, **GCP Professional Cloud Architect**

Cloud & DevOps: GCP, Vertex AI, Docker, Kubernetes (GKE), Azure Pipelines, GitHub, Rancher, Splunk

Web & Databases: FastAPI, Next.js, React, Redis, MySQL, PostgreSQL, MongoDB

Languages: Python, JavaScript, C++

Experience

Rocket, India (formerly Mr. Cooper) Jul 2024 - Present

Software Engineer I

- **Decision Graph (Modular Routing):** Replaced a brittle single-prompt LLM classifier with a modular **decision graph**, lifting reviewer-agreed accuracy from **29% to 77% (2.7x)** and cutting wrong classifications by **95%** across ∼**17,500** weekly inbound items. **Auto-closes with 100% accuracy and zero false positives.**
  - The question is split into smaller focused prompts; several models answer independently, plain code settles disagreements, and a stronger oversight model vetoes anything unsafe before it runs.
  - **HITL** fallback routes low-confidence cases to human review, while new versions ship as a **shadow deployment** running beside live traffic — a single flag cuts over, or rolls back, instantly with no deploy.
- **In A Box (Multi-Agent System):** Built and deployed a scalable multi-agent workflow to automate complex multi-document verification across **50+ checkpoints**, achieving a **67% reduction** in manual processing effort.
  - Built a large-scale orchestration graph using **LangGraph** for Loan Modification workflows.
  - Integrated multiple enterprise systems (**FileNet, LSAMS, CMOD)** to enable reliable data aggregation.
  - Implemented cross-verification pipelines across **5+ document types**, combining classification, extraction, and validation within a unified graph.
- **PyroAI Docs (GenAI Extraction):** Built a production-grade LLM-based document extraction system, improving accuracy from **9% to 90%** and enabling straight-through processing across **15+ financial document types**, handling **2000+ pages/day**.
  - Migrated legacy Deep Learning-based pipelines to **LLM**-driven architectures and engineered optimized prompt pipelines for diverse document formats and edge cases, significantly improving robustness and maintainability.
  - Reduced monthly cloud infrastructure costs from **$30K+ to < $10K** while decreasing dependency on human-in-the-loop (**HITL**) operations, improving overall system efficiency.
- **AI Guardian (LLM Validation):** Led the development of a scalable **LLM evaluation and validation framework** to benchmark production models against external competitors. Built backend services using **FastAPI, Docker, GKE**.
  - Enabled systematic performance tracking, evaluation, and model comparison across multiple dimensions using **LLM-as-judge**, **Cosine Similarity** and **Levenshtein Distance** metrics.

Education

MEPCO Schlenk Engineering College, Tamil Nadu, India 2020 - 2024

B.Tech in **Artificial Intelligence and Data Science** ***CGPA: 8.49/10***

Relevant Coursework: Machine and Deep Learning, Generative AI, Data Science, IoT, Data Analytics, Cloud Computing.

Awards and Recognition

- **The Rookie (Mr. Cooper, 2025):** Recognized as the top-performing new hire (**Top 1%**) in the PyroAI team for 2024-25.
- **IEEE Conference Publication (2024):** Published "**Pressure, Queue, and Average Speed**-Based **Multi-Agent DQN**" demonstrating superior traffic flow and reduced congestion against fixed-time schedules using **3-tier reward function**.
- **Best Outstanding Student (2024):** Selected as the **sole** recipient out of **50** peers (**Top 2%**) for maintaining an **8.49 CGPA** while **directing** events and operations for the Microsoft Campus Club and Consumer Club.